A four-point internal orbit gives a nonterminal separator

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le \mathrm{Alt}(\Omega)$ be transitive with a connected string C-marking and all maximal omissions intransitive. Assume $n = 2r+1$ and the actual six-point head on $D = \{1, \dots, 6\}$, with

$$\tau = \rho_3 = (6\ b)\delta, \qquad R = G_{\ge 3} = \mathrm{Alt}(\Omega \setminus \{1, \dots, 5\}).$$

The involution δ fixes D, b; $\sigma = \rho_4$ fixes D; $H = G_{\ge 5}$ fixes D, b. Put $K = G_{>5}$. The actual pure-overlap argument gives $x = b^\sigma \ne b$, $\delta x = x$ and $Kx = x$.

Suppose H has precisely the seven fixed points $D \cup \{b\}$ and three nontrivial orbits E, W, V. Retain the following literal assignment data:

- $|E| = 2$, $|W| = 4$, and $|V|$ is even. The involution δ swaps E, is free on W, and fixes V.
- The normalized assigned sets are $F_E = \varnothing$, $F_W = \{j\}$ for some $j \ge 6$, and $F_V = \{5, \dots, r-1\} \setminus \{j\}$. Each assigned colour has two full ambient-eligible edges in its assigned orbit.
- K is intransitive on W.

Theorem 1. *The index j is strictly below $r-1$, and*

$$|\operatorname{Fix}(G_{>j})| \ge 2j + 2.$$

Thus this precise four-point-orbit budget has a nonterminal local separator. No assertion about its transfer through a subsequent external attachment is made.

Proof. The two pairs of $\delta|_W$ are preserved by H. An eligible j-edge cannot be one of these pairs, because $\tau \in G_j$ joins that pair. Thus ρ_j exchanges the pair vertices. Every other mark of H has trivial pair image: if one also exchanged them, it and τ would connect all four points inside G_j, contradicting the eligible edge.

The group K contains ρ_j, so it is transitive on the two pair vertices. As it is intransitive on W, it has exactly two orbits of size two, each meeting each pair once. A pair-trivial element of K fixes both orbits and hence all four points. Consequently $K|_W$ is the diagonal S_2 on these two rows, $\rho_j|_W$ is a double transposition, and every other mark of K fixes W pointwise. This proves the needed two-pair equality directly; no theorem stated only for at least three pairs is imported.

Write $Q = \langle \rho_j \rangle$ and $Z_\ell = W^{G_{\le \ell}}$ for $4 \le \ell \le j-2$, where the superscript denotes the union of all translates. The prefix $G_{\le 4}$ centralizes K, so Z_4 is a union of its two-point row orbits. It contains at least three such rows. Indeed, if $Z_4 = W$, both $G_{\le 4}$ and H would preserve W, contradicting transitivity of G.

Every Z_ℓ is a union of nontrivial two-point Q-orbits: $G_{\le \ell}$ centralizes Q in the stated range. These sets exclude the Q-fixed head and are therefore proper. For $4 \le \ell \le j-3$, each mark of index at least $\ell + 2$, other than ρ_j, fixes W and commutes with $G_{\le \ell}$, so fixes Z_ℓ pointwise. The mark ρ_j preserves it, as does $G_{\le \ell}$. Thus $\rho_{\ell+1}$ cannot preserve Z_ℓ, or the whole group would preserve a proper set. It adds at least one complete two-point Q-orbit. Hence

$$|Z_{j-2}| \ge 6 + 2(j-6) = 2j - 6. \tag{1}$$

Every mark after j fixes W and commutes with $G_{\le j-2}$, so fixes Z_{j-2}. It also fixes the eight distinct points $D \cup \{b, x\}$, since it belongs to K. These points are fixed by Q and therefore disjoint from Z_{j-2}. Adding them to (1) proves the required bound.

Suppose now that $j = r - 1$, so $n = 2j + 3$. Let C_k be the actual $G_{\le k}$-orbit containing D. The degree-independent support-merging proof of the acquired prefix-kernel theorem gives

$$\mathrm{Alt}(C_k) \le G_{\le k}, \qquad |C_k| \ge k + 4 \quad (k \ge 3).$$

Here is the precise part used. The literal head gives pure $\mathrm{Alt}(D \cup \{b\})$ and then pure $\mathrm{Alt}(D \cup \{b, x\})$. Each subsequent mark fixes C_{k-2} by string commutation and its pure alternating kernel. Merging the overlapping conjugate supports supplies pure $\mathrm{Alt}(C_k)$. If a head orbit failed to grow, every later mark would fix it by the same commutation, contradicting transitivity or independence. These arguments use neither $n = 2r$ nor an alternating rank induction.

Since Q centralizes $G_{\le j-2}$, it fixes C_{j-2} pointwise. This set is disjoint from Z_{j-2}, all of whose points are moved by Q. Consequently

$$2j + 3 = n \ge (2j - 6) + (j + 2) = 3j - 4,$$

so $j \le 7$. There are only two possibilities. If $j = 6$, then $r = 7$, $n = 15$ and $|V| = n - 7 - 2 - 4 = 2$. But $F_V = \{5\}$ requires two disjoint eligible edges on this two-point set, which is impossible. If $j = 7$, then $r = 8$, $n = 17$, $|V| = 4$ and $F_V = \{5, 6\}$. The two eligible 6-edges on V make $\rho_6|_V$ fixed-point-free. On the other hand $x \in V$, because it is outside D, b and is fixed by δ, which is free on $E \cup W$. The group K fixes x, and contains ρ_6. This is a contradiction. Thus $j < r - 1$. □

Attribution and boundary. The row-transport and prefix-growth strategy is acquired; the two-pair equality, its bound $2j + 2$, and the terminal labels six and seven are proved here. The pure alternating kernel is the acquired support-merging argument, retaining the root collaborator's initial kernel contribution and all complete predecessor costs. The three-orbit assignment budget is explicit input; neither its necessity in every seven-fixed-point configuration nor global separator transfer is asserted. No new computation, formal theorem, original or question credit is claimed.